Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/04/2005
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Enhanced Labeling Techniques to Study the Cytoskeleton During Root Growth and Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	09/01/2001	**End Date:**	03/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	An important function of the cytoskeleton lies in the regulation of plant form and orientation. Since gravity affects plant form, it is likely that the plant cytoskeleton is a target of gravity-induced modifications in plant morphogenesis. This proposal will use roots to study the dynamic behavior of the actin and microtubule cytoskeleton during gravity-induced differential growth in plants. Roots are ideal subjects for studies on the mechanisms of growth control because of the clear distinctions between developmental zones along the longitudinal axis (e.g. meristem, elongation and maturation zones) and because roots have a predictable growth response to gravity (i.e. curvature). The ability to observe these developmental stages in a single root and the predictable gravity-induced growth responses has allowed correlative studies between a specific type of cytoskeletal structure and the stage of cellular development. The proposed research will re-examine the actin cytoskeleton in gravistimulated roots using improved labeling methods that include green fluorescent protein (GFP)-cytoskeletal constructs for live cell imaging and an MBS crosslinking/glycerol permeabilization method for fixed roots. These approaches will be combined with classical hormone transport studies, computer based growth measurements and ion imaging using fluorescent sensors. Our experiments will be applied to roots of various species including Arabidopsis thaliana, Zea mays, Linum usitatissimum and Nicotiana benthamiana but will also employ Medicago truncatula because of the expanding genomic resources available at the Noble Foundation for this plant species. The objective of this proposal is to develop improved labeling methodology to help in our understanding of the function of the cytoskeleton during root growth and gravity responses. The results from this research will aid in developing a mechanistic knowledge on how gravity affects plant development at the cellular and molecular level. The cytoskeleton in particular is an attractive cellular system to investigate because of its importance in several fundamental cellular processes such as the the establishment of plant form and orientation. These traits are critical for the targeted improvement of plants not only to increase yields for food production but also to modify plant stature to reduce biomass in an advance life support system needed for extended space missions. Through this research, we have also provided unique educational opportunities to students at all levels. Educational and otureach programs at the Noble Foundation have provided bio-imaging resources for undergraduate/high school student interships and teacher training.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective of this research is to investigate the role of the cytoskeleton in root responses to gravity. Results from the proposed research will extend our understanding of how plants tell their correct orientation (i.e. know up from down). The ability of plants to respond to gravity is important for agriculture since crop productivity depends largely on plant form so it can more efficiently explore the soil for nutrients and water as well as optimize light absorption for photosynthesis. The cytoskeleton is critical to achieving the correct orientation of the plant thus the basic knowledge gained from this research should be helpful for optimizing strategies for crop production on Earth.
Task Progress:	Nine (9) peer reviewed publications have resulted from this research with four peer-reviewed articles published during this reporting period. From this research we have shown that the actin cytoskeleton is likely to have a role in regulating plant gravitropism that is different from existing models. A popular explanation for actin's involvement in plant gravity sensing and signaling is that sedimenting amyloplasts pull on the actin cytoskeleton leading to the activation of channels, which then results in the influx of ions into the cytoplasm initiating a chain of events leading to organ bending. Our work has shown that this model may need revision because instead of being inhibited, the sensitivity and response of roots to gravity is significantly increased when the actin cytoskeleton is disrupted with Latrunculin B (Lat B). The enhanced gravity response was most pronounced when Lat B-treated roots were given a brief period of horizontal stimulation (ca. 10 min) followed by extended periods of rotation on a clinostat to randomize the gravity vector. Under such conditions, Lat B-treated roots would continue to curve strongly in the direction of the original gravity vector while control roots would bend weakly and in proportion to time of horizontal stimulation. These observations led us to propose that disrupting F-actin may be affecting the reset mechanisms that prevent the root from overshooting the vertical. Alternatively, these results could indicate that Lat B-treated roots remember the initial gravity stimulus despite randomizing the gravity signal with a clinostat. We conducted studies using seedling roots of Arabidopsis so we could exploit the molecular tools available in this model plant. We studied cellular events that are well known to be associated with root gravity responses including the development of asymmetric auxin gradients across opposite flanks of the root, cytoplasmic pH signaling in the gravity sensing columella cells of the root cap, and amyloplast dynamics and sedimentation in the columella cells. Our results show that all three of these events were modified by Lat B-treatment indicating that the actin cytoskeleton is involved in modulating these events. For example, indirect visualization of auxin redistribution using DR5::GUS, an auxin responsive reporter, revealed that the stronger curvature response of roots upon disrupting the actin network was accompanied by a persistent lateral auxin gradient. Actin disruption also increased the rate of amyloplast sedimentation in the columella and extended the duration of the gravity-induced pH changes in the cap, both of which are linked to the initial events of gravity perception and signaling. The amplification of these signals when actin was disrupted with Lat B indicates that the actin cytoskeleton functions by downregulating components of the gravity signal transduction chain. In this reporting period, we have also generated a novel fluorescent protein fusion that allows improved labeling of actin dynamics and organization in living plant cells. This reporter is based on GFP fusions to Arabidopsis fimbrin 1 (AtFim1), an F-actin cross-linking protein that contains two actin binding domains (ABDs) arranged in tandem within a single polypeptide chain. We showed that GFP fusions to truncated variants of the AtFim1 protein, specifically ABD2, can decorate F-actin in living cells and allow the visualization of a finer and more dynamic F-actin network compared to the more popular Talin-GFP actin reporter . More importantly, stable expression of these Fimbrin based GFP reporters in Arabidopsis allowed imaging of F-actin in different developmental regions of the root including the root cap, which is important for our studies on gravitropism. Also, we have generated tobacco BY2 cells and hairy roots of Medicago truncatula stably expressing these newly developed actin markers paving the way for detailed studies of actin turn over and dynamics in other important plant biological systems. Since we first published on this new F-actin reporter gene, we have had numerous requests for this construct from various researchers in the plant science community.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Articles in Peer-reviewed Journals	Hou, G., Hill, J.P., Blancaflor, E.B. "Developmental anatomy and auxin response of lateral root formation in Ceratopteris richardii" Journal of Experimental Botany 397: 685-693 , Mar-2004

Articles in Peer-reviewed Journals	Wang, Y.-S., Motes, C.M., Mohamalawari, D.R., Blancaflor, E.B. "Green fluorescent protein fusions to Arabidopsis fimbrin 1 for spatio-temporal imaging of F-actin dynamics in roots" Cell Motility and the Cytoskeleton 59: 79-93 , Oct-2004
Articles in Peer-reviewed Journals	Hou, G., Kramer, V.L., Wang, Y.-S., Chen, R., Perbal, G., Gilroy, S., Blancaflor, E.B. "The promotion of gravitropism in Arabidopsis roots upon actin disruption is coupled with the extended alkalinization of the columella cytoplasm and a persistent lateral auxin gradient" Plant Journal 31: 113-125 , Jul-2004
Articles in Peer-reviewed Journals	Blancaflor, E.B., Masson, P.H. "Plant gravitropism. Unraveling the ups and downs of a complex process" Plant Physiology 113: 1677-1690 , Dec-2003
Journal/Magazine covers	Hou, G., Kramer, V.L., Wang, Y.-S., Chen, R., Perbal, G., Gilroy, S., Blancaflor, E.B. Plant Journal Plant Journal 31: 113-125 , Jul-2004
Papers from Meeting Proceedings	Wang, Y.S., Motes, C.M., Blancaflor, E.B. "Spatio-temporal imaging of F-actin dynamics in developing roots using green fluorescent protein fusions to Arabidopsis Fimbrin 1" American Society of Plant Biologists Annual Meeting Orlando, Florida Jul-2004
Papers from Meeting Proceedings	Blancaflor, E.B., Hou, G., Kramer, V.L, Wang, Y.-S., Chen, R., Gilroy, S. "The actin cytoskeleton is a negative regulator of root gravitropism " American Society of Plant Biologists Annual Meeting Orlando, Florida Jul-2004
Papers from Meeting Proceedings	Hou, G., Kramer, V.L., Wang, Y.-S., Chen, R., Perbal, G., Gilroy, S., Blancaflor, E.B. "The promotion of gravitropism in Arabidopsis roots upon actin disruption is coupled with the extended alkalinization of the columella cytoplasm and a persistent lateral auxin gradient" American Society of Plant Biologists Southern Section Meeting, Lafayette, Louisiana Mar-2004
Papers from Meeting Proceedings	Wang, Y.-S., Motes, C.M., Mohamalawari, D.R., Blancaflor, E.B. "Green fluorescent protein fusions to Arabidopsis fimbrin 1 for spatio-temporal imaging of F-actin dynamics in roots" American Society of Plant Biologists Southern Section Meeting, Lafayette, Louisiana Mar-2004
Papers from Meeting Proceedings	Yoo, C.M., Wang, Y.S., Blancaflor, E.B. "Unraveling the Role of the Actin Cytoskeleton in Root Growth and Gravitropism Using Green Fluorescent Protein (GFP) Reporters and Knockouts to Actin Regulatory Proteins" American Society for Gravitational and Space Biology Annual Meeting, New York, New York Nov-2004
Papers from Meeting Proceedings	Blancaflor, E.B., Wang, Y.-S., Motes, C.M. "Spatio-Temporal Imaging of F-actin Dynamics in Plants Using Green Fluorescent Protein Fusions to Arabidopsis Fimbrin 1" American Society for Gravitational and Space Biology Annual Meeting, New York, NY Nov-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravitropism" Biology Department Seminar Series, Texas A.M. University, College Station Texas Dec-2003
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravity responses. Cytoskeletal and ionic signaling during root gravitropism" Botany Department Seminar Series, Oklahoma State University, Stillwater, Oklahoma Mar-2004
Presentation	Blancaflor, E.B. "The actin cytoskeleton and its role in plant gravity responses" 15th Symposium in Plant Physiology, Pennsylvania State University, State College, Pennsylvania May-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravity responses" Botany Department Seminar Series, University of Oklahoma, Norman, Oklahoma Sep-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravity responses. Cytoskeletal and ion signaling in the root cap" Department of Environmental and Plant Biology Seminar Series, Ohio University, Athens, Ohio Oct-2004
Presentation	Blancaflor, E.B. "Cytoskeletal and lipid dependent regulation of seedling development in Arabidopsis" School of Biological Sciences Seminar Series, Washington State University, Pullman, Washington Nov-2004